# 非線形LUT補間（自作関数）

2025/09/23

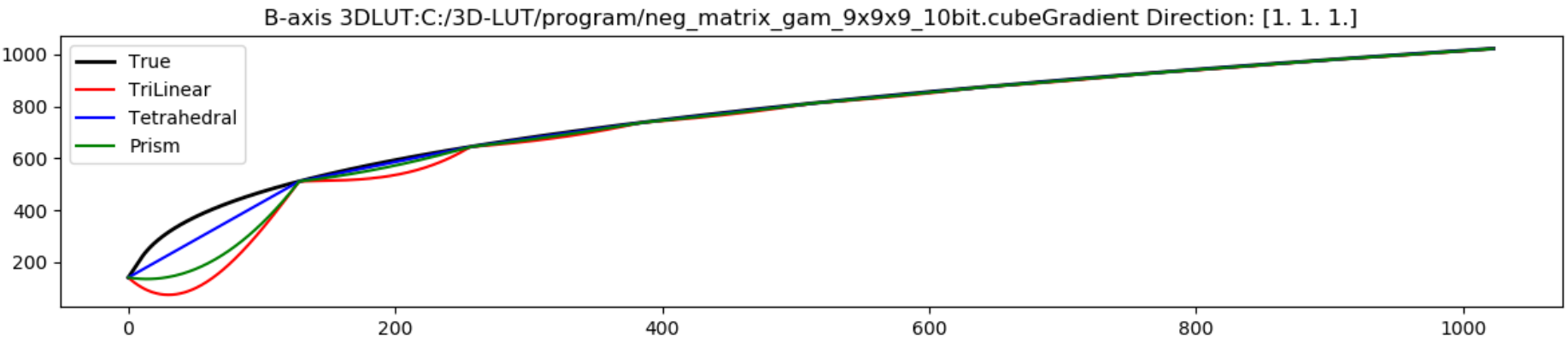
make_non_linear_LUT.py

grad_using_3dlut.py

```
# より適切な閾値を設定（ノイズ低減のため）
threshold = 0.008856  # (6/29)^3 より大きな値でスムーズな変換

X = 2*r1 - 0.5*g1 - 0.5*b1

# 数値安定性を向上させた立方根計算
if X > threshold:
    rr = (X)**(1/3)
elif abs(X) <= threshold:
    # 線形近似（連続性を保証）
    # f(x) = x^(1/3) の x=threshold での微分値は 1/(3*threshold^(2/3))
    rr = X / (3 * (threshold**(2/3))) + threshold**(1/3) - threshold / (3 * (threshold**(2/3)))
else:
    rr = -((-X)**(1/3))
```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TriLinear
Tetrahedral
Prism
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800
600
400
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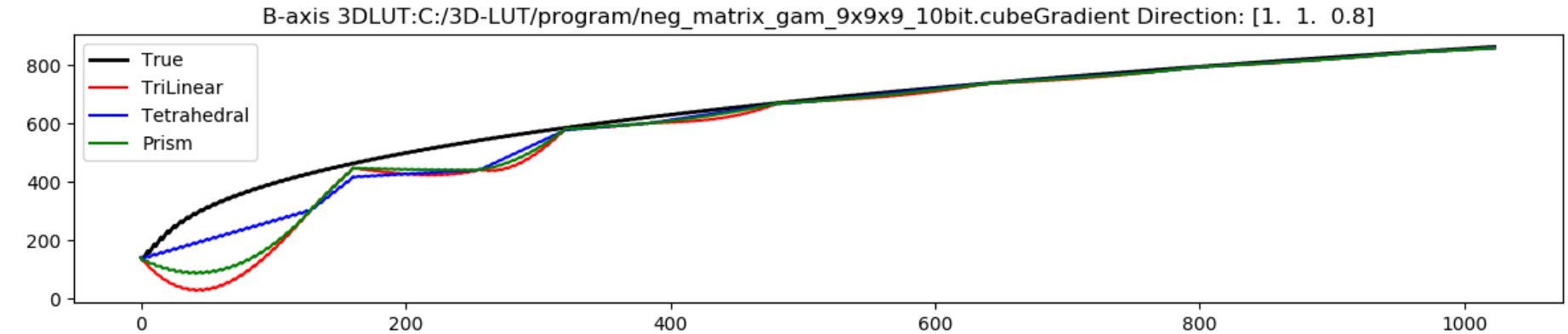
800
1000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Tetrahedral
Prism
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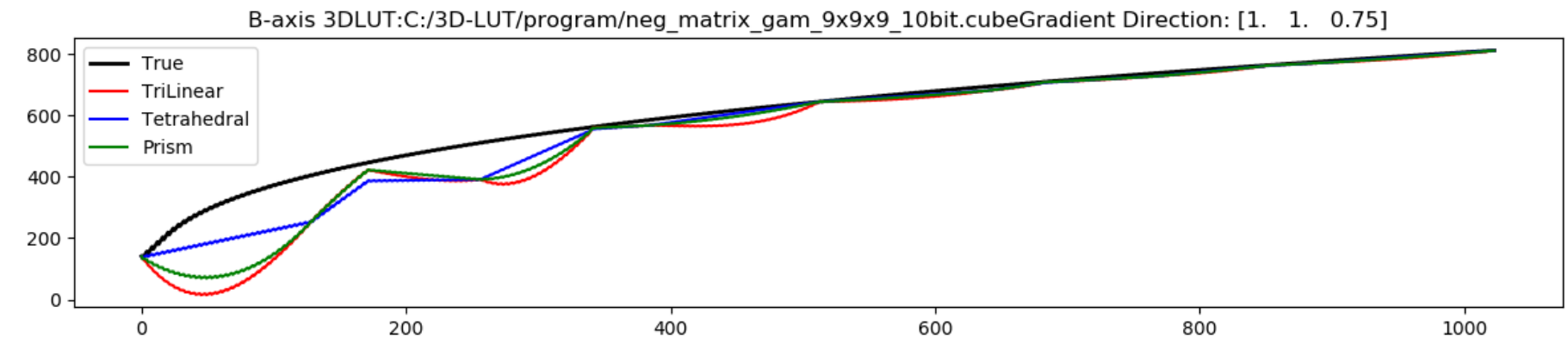
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Tetrahedral
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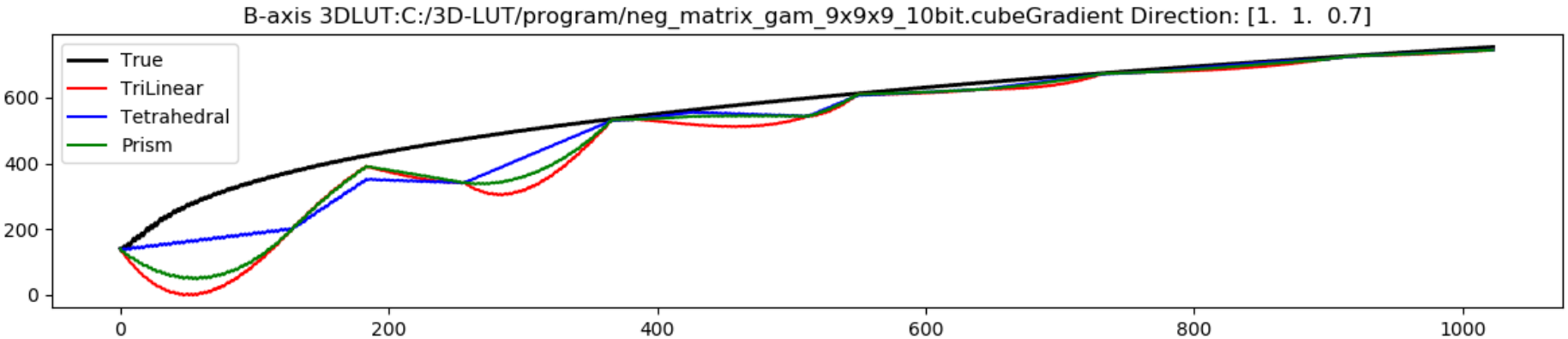
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Tetrahedral
Prism
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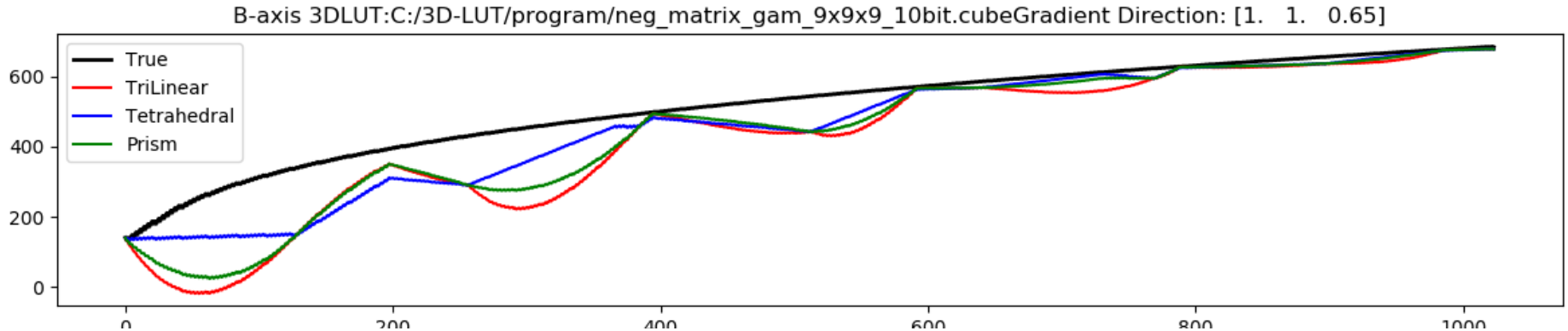
400
600
800
1000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TriLinear
Tetrahedral
Prism
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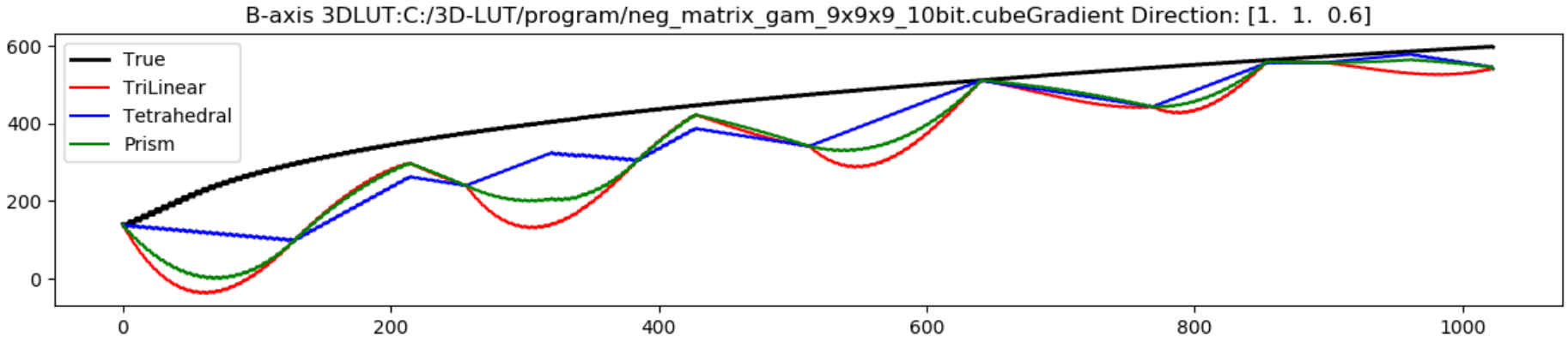
400
600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TriLinear
Tetrahedral
Prism
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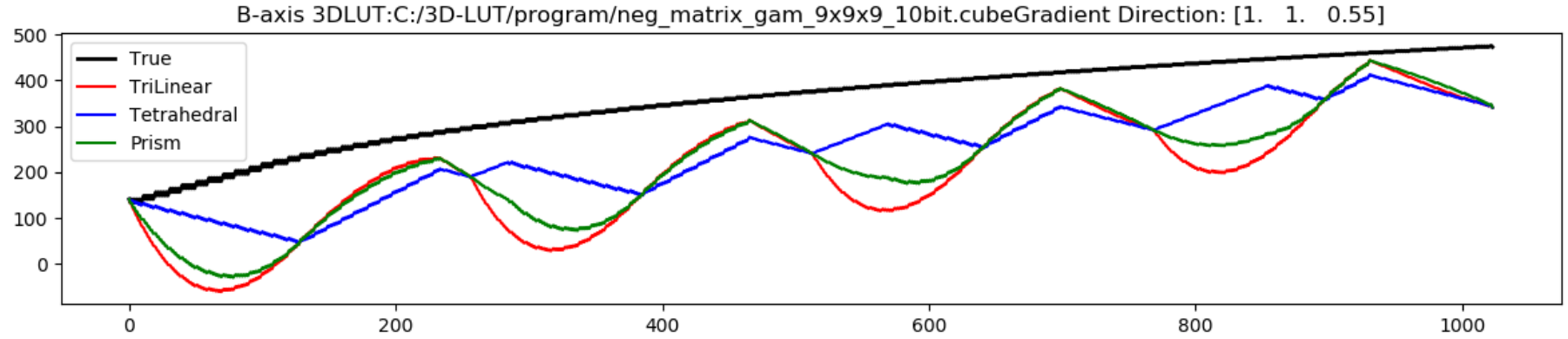
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TriLinear
Tetrahedral
Prism
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800
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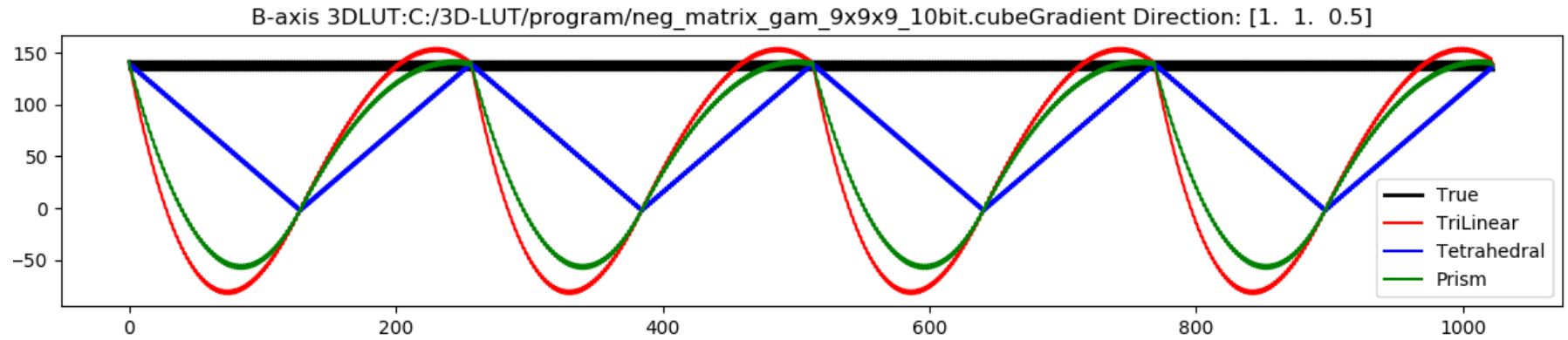
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Tetrahedral
Prism
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800
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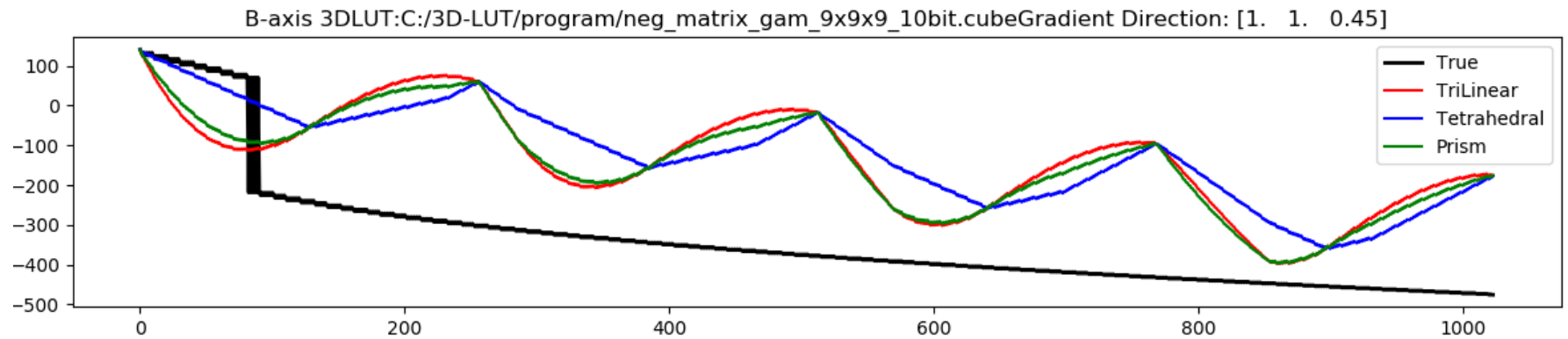
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TriLinear
Tetrahedral
Prism
100
0
−100
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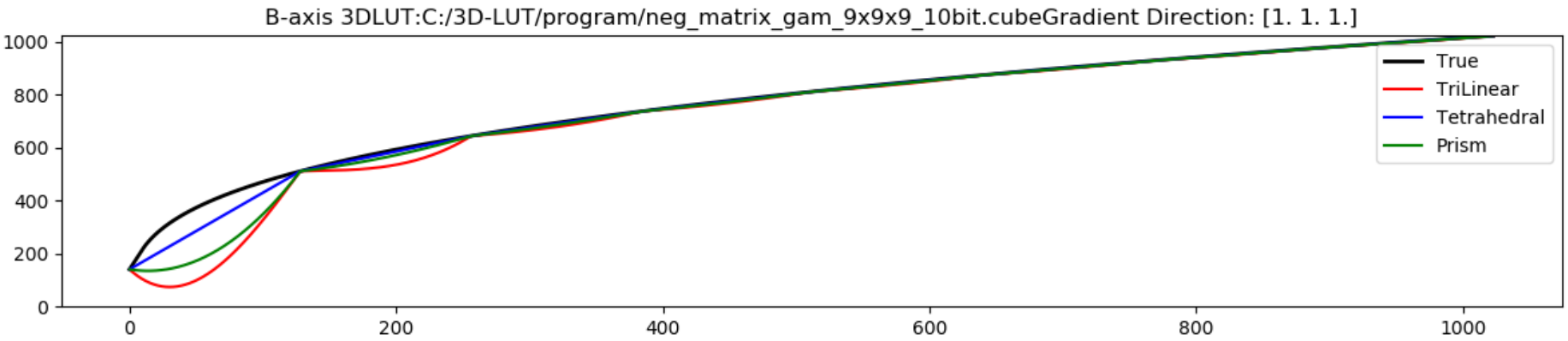
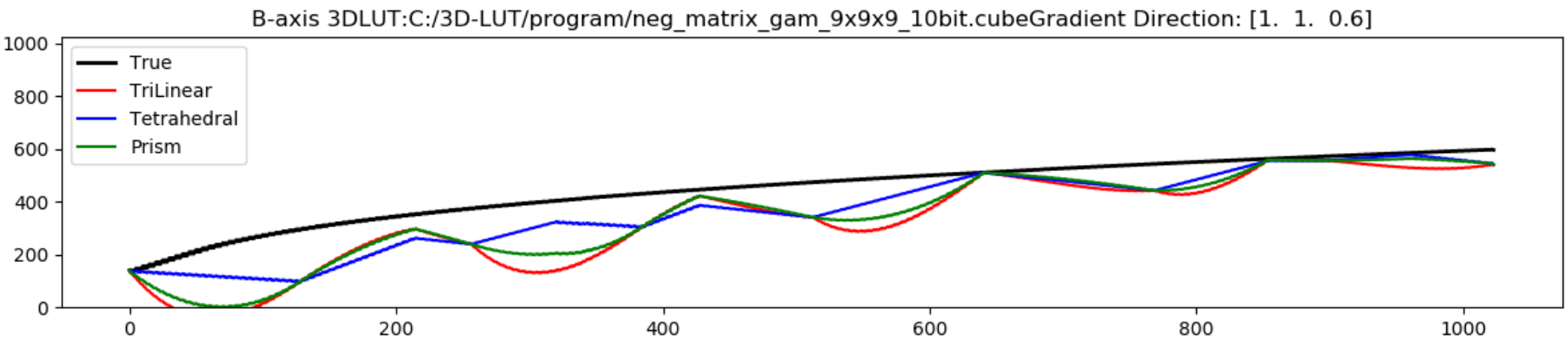
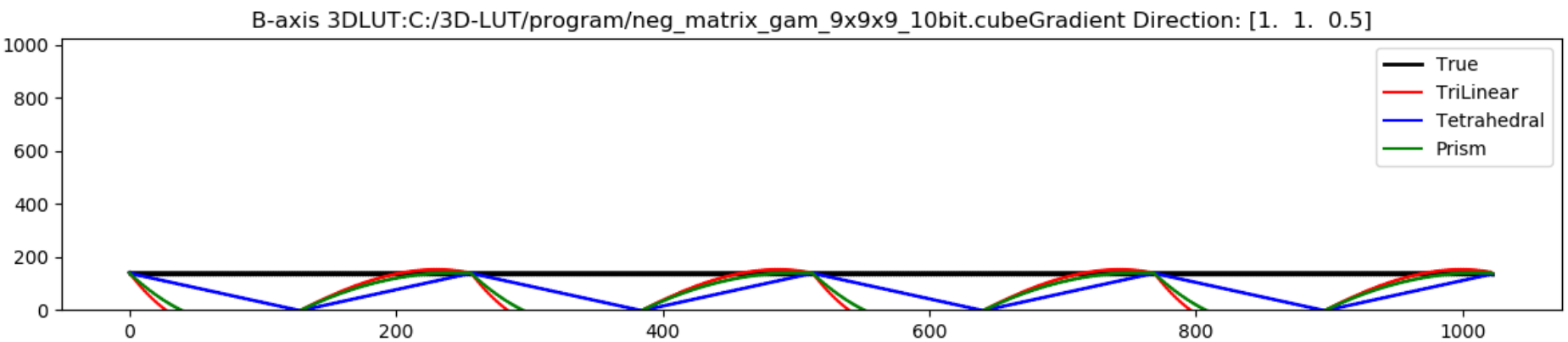
−400
−500
0
200
400
600
800
1000

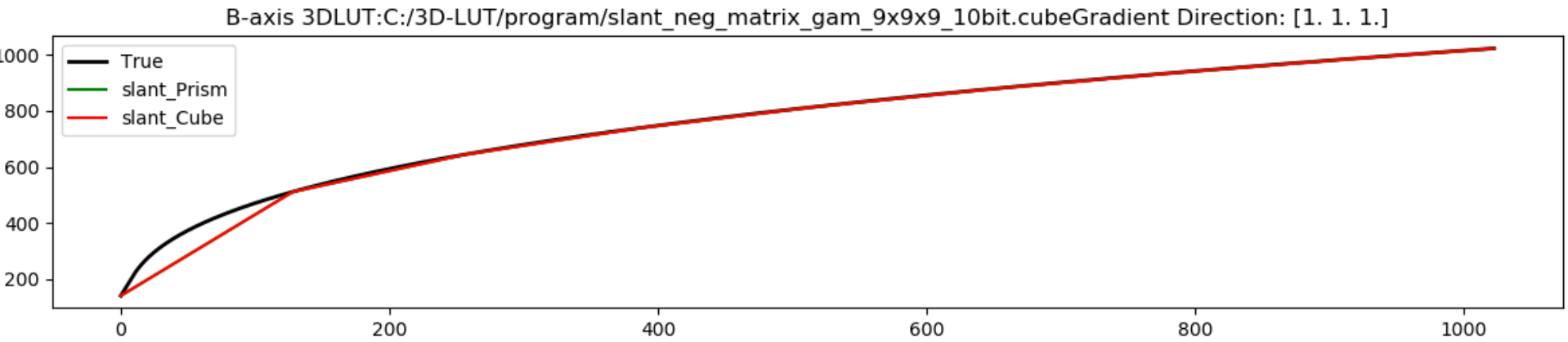
# 縦軸を同一レンジで観測してみる



# SLANT系の補間


B-axis 3DLUT:C:/3D-LUT/program/slant_neg_matrix_gam_9x9x9_10bit.cubeGradient Direction: [1. 1. 1.]
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slant_Prism
slant_Cube
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800
600
400
200
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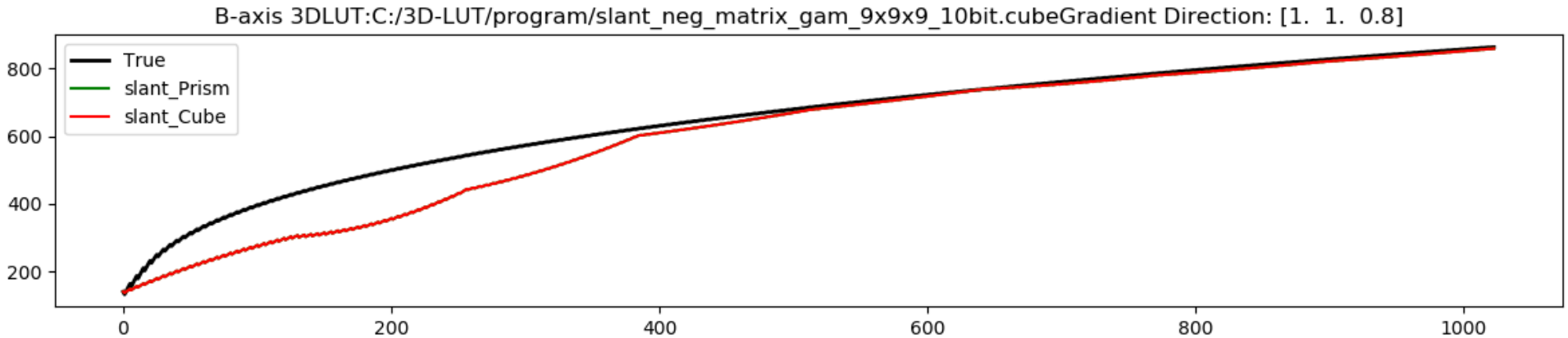
200
400
600
800
1000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slant_Cube
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400
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400
600
800
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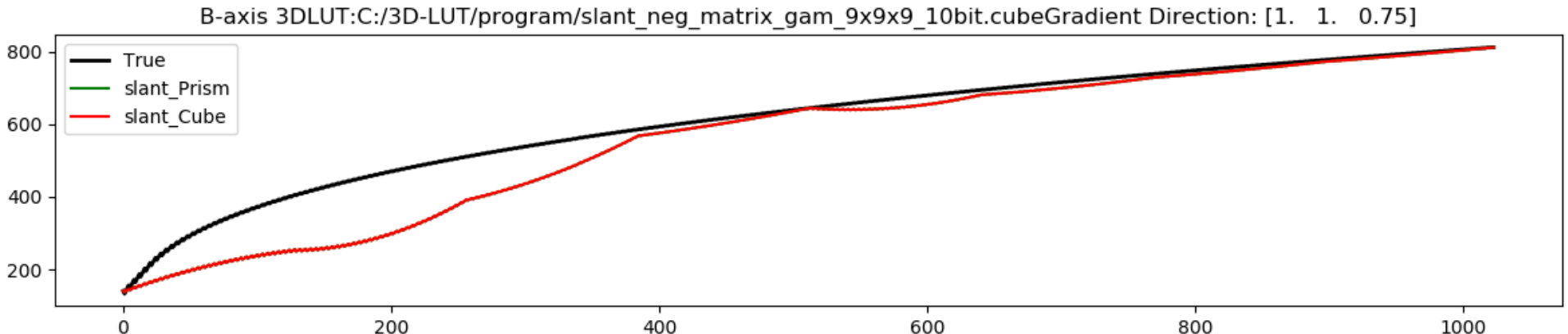
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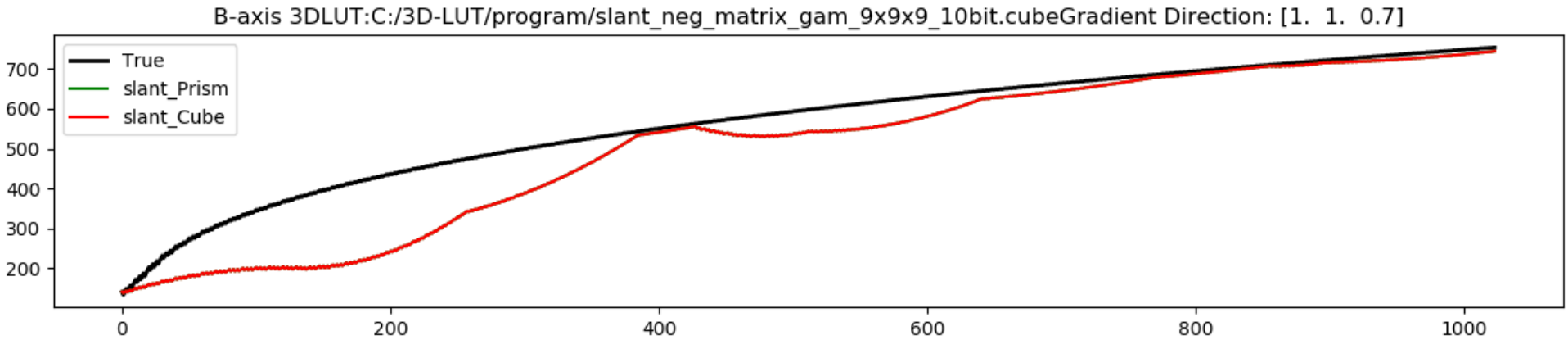
True
slant_Prism
slant_Cube
800
600
400
200
0
200
400
600
800
1000

# SLANT系の補間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slant_Prism
slant_Cube
700
600
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400
300
200
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200
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800
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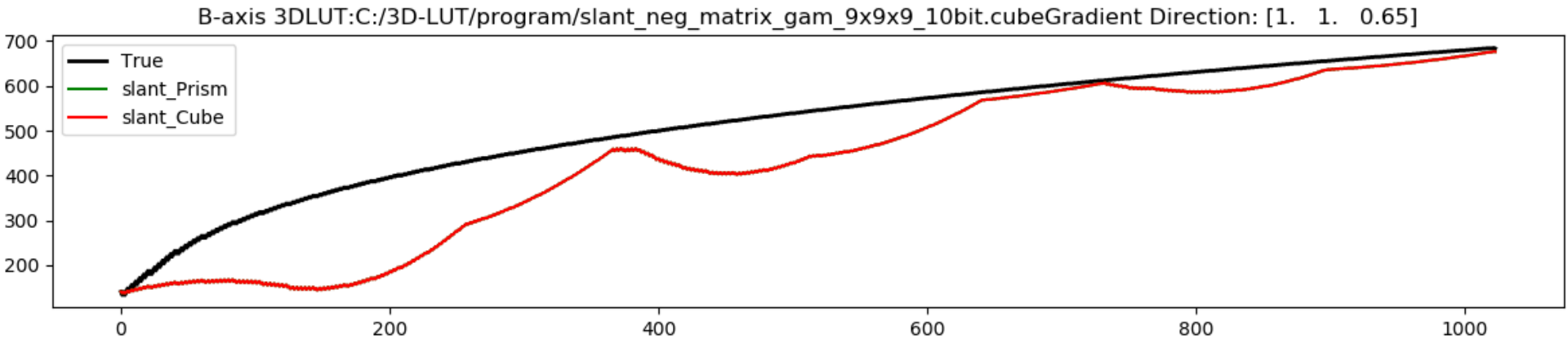
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slant_Prism
slant_Cube
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600
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400
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200
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200
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800
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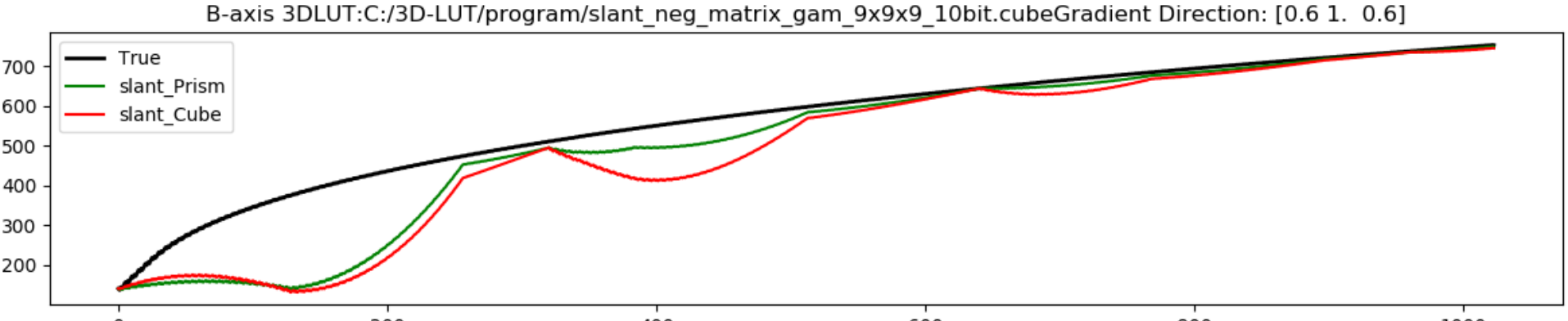
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slant_Prism
slant_Cube
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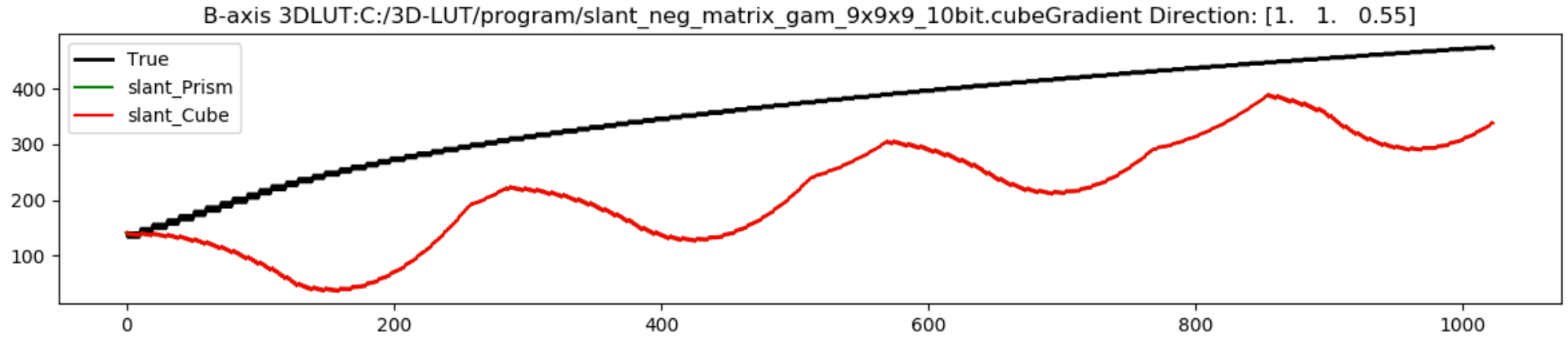
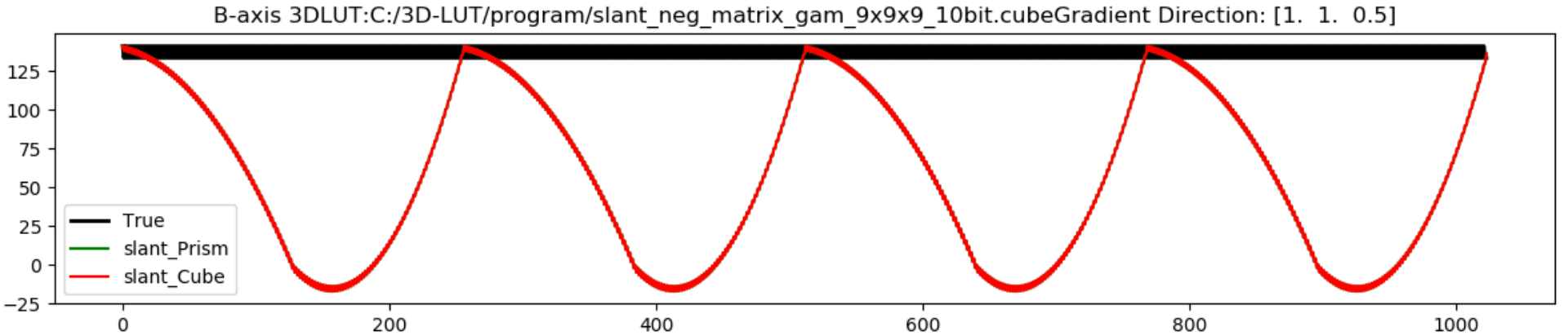
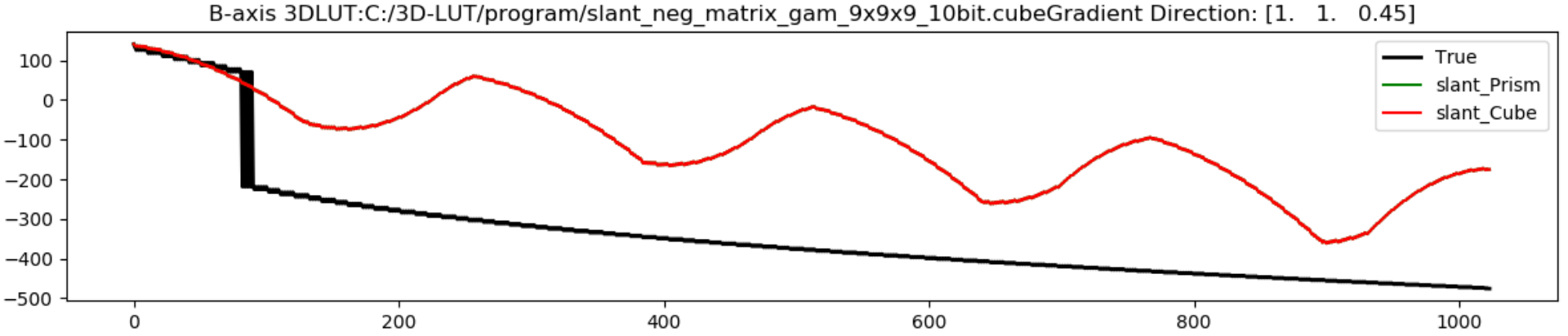
600
500
400
300
200

# SLANT系の補間


B-axis 3DLUT:C:/3D-LUT/program/slant_neg_matrix_gam_9x9x9_10bit.cubeGradient Direction: [1. 1. 0.55]
True
slant_Prism
slant_Cube
400
300
200
100
0
200
400
600
800
1000


B-axis 3DLUT:C:/3D-LUT/program/slant_neg_matrix_gam_9x9x9_10bit.cubeGradient Direction: [1. 1. 0.5]
True
slant_Prism
slant_Cube
125
100
75
50
25
0
−25
0
200
400
600
800
1000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slant_Prism
slant_Cube
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0
−100
−200
−300
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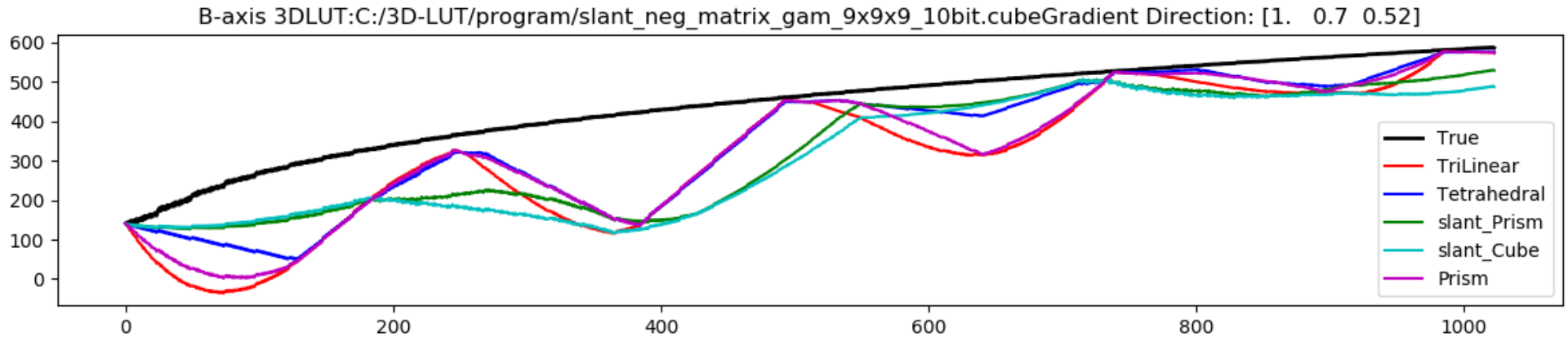
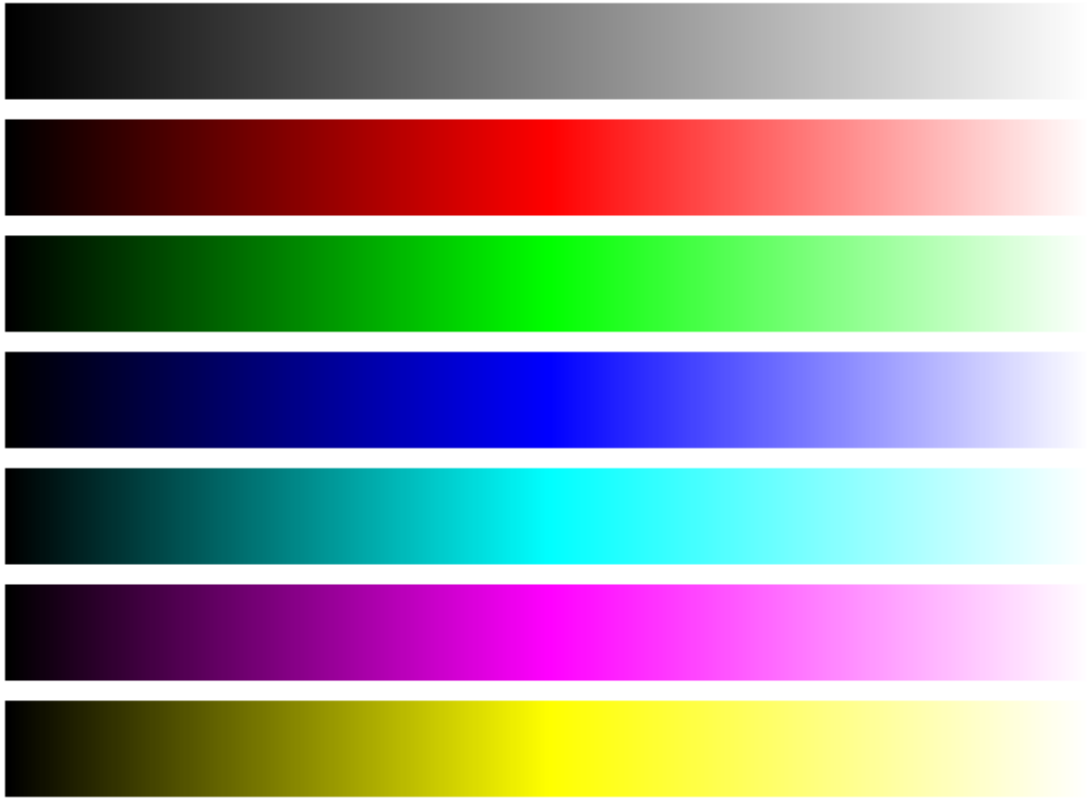
−500
0
200
400
600
800
1000

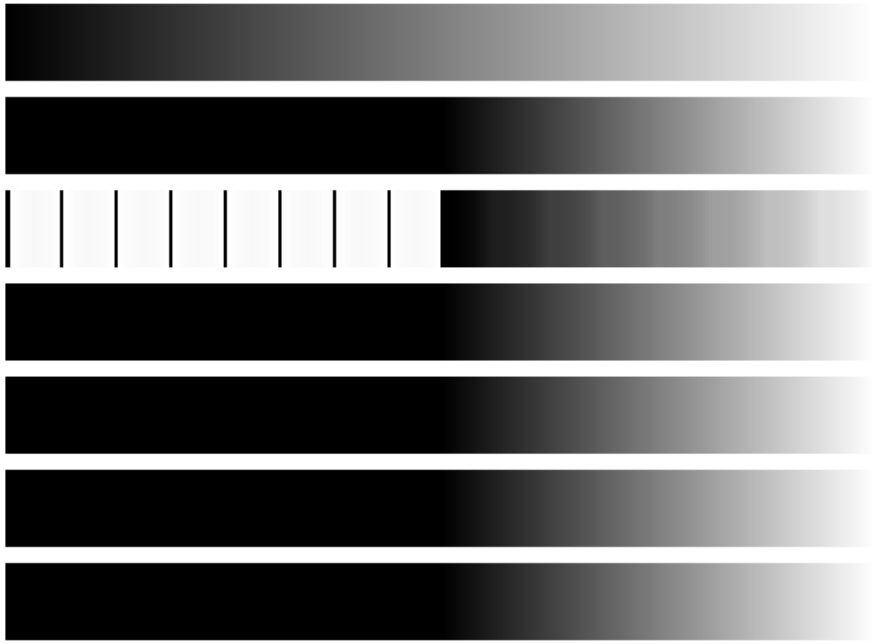
# SLANT系3D-LUT　通常３D-LUT　の補間比較





# SLANT_CUBE vs SLANT_PRISM

## slant_cube



## slant_prism

BUGではないみたい。。。
slant-3DLUTの張り出し部分の蓄積値が(-)
になっている

## imgproc_using-slant3dlut.py

"slant_min_9x9x9_8bit.cube"

# SLANT_CUBE vs SLANT_PRISM

slant_cube

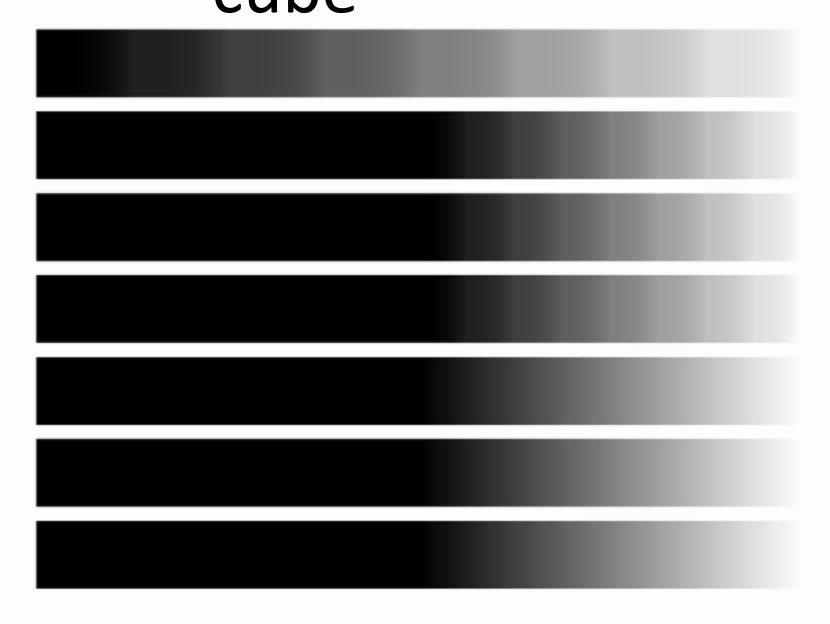
slant_prism

imgproc_using-slant3dlut.py

"slant_min_9x9x9_8bit.cube"

# _CUBE vs _PRISM

cube



prism

tetra